

Assignment: Data and Signals

1. (a) The period of a signal is 50 ms. Calculate its frequency in Hz.
(b) If the frequency of a sine wave is 200 Hz, what is its period?
2. A sine wave has an amplitude of 10 V and a frequency of 500 Hz.
(a) What is the angular frequency (ω)?
(b) Draw one complete cycle of the waveform.
3. A periodic signal is composed of sine waves with frequencies 100 Hz, 300 Hz, 500 Hz, and 700 Hz.
(a) Find the bandwidth of the signal.
(b) Draw the frequency spectrum.
4. A digital signal uses 16 distinct levels.
(a) How many bits per level are required?
(b) If the bit rate is 1 Mbps, what is the minimum bandwidth required for baseband transmission?
5. A signal with power 10 mW is attenuated to 2.5 mW after transmission.
(a) Calculate the attenuation in dB.
(b) Interpret the result in terms of signal quality.
6. A nonperiodic composite signal has a bandwidth of 300 kHz with a middle frequency of 1 MHz.
(a) Find the highest and lowest frequencies.
(b) Sketch the frequency spectrum.
7. A digital signal with 8 levels is transmitted at a baud rate of 2400.
(a) How many bits per baud are transmitted?
(b) Calculate the bit rate.
8. Define the following terms:
(a) Amplitude
(b) Frequency
(c) Phase
9. Differentiate between analog and digital signals with an example of each.
10. Explain how attenuation, distortion, and noise affect signal transmission. Provide a real-world example
11. A data stream 101100 is encoded using **NRZ-L** with +5 V for 1 and -5 V for 0.
12. Draw the waveform and calculate the average signal power if each bit lasts 1 μ s.
13. A sequence 1001110 is transmitted using **NRZ-I**. Assume the signal starts at +5 V. Write the voltage sequence.
14. In **Manchester encoding**, the bitstream 11001 is sent.
 - Show the waveform transitions.
 - How many transitions occur in total?

15. In an **ASK system**, carrier amplitude for 0 is 0 V and for 1 is 5 V. If the bit rate is 2 kbps, what is the minimum bandwidth required?
16. **Error Detection (CRC)**
Given the dataword 101101 and the generator polynomial $G(x) = x^3 + x + 1$, compute:
a) The codeword transmitted.
b) Show how the receiver detects an error if the received word is 101101101.
17. **Hamming Code**
A 7-bit Hamming code is used for error detection and correction. The message is 1011.
a) Construct the Hamming code.
b) If the received codeword is 1010110, detect and correct the error.
18. **Go-Back-N ARQ**
A sender uses Go-Back-N ARQ with a window size of 4 and sequence numbers ranging from 0–7.
a) Illustrate the transmission timeline if frame 2 is lost and NAK is received.
b) Show the sequence of retransmissions.
19. **Stop-and-Wait ARQ**
If the bit rate is 1 Mbps, propagation delay = 20 ms, and frame size = 1000 bits:
a) Calculate the utilization of Stop-and-Wait protocol.
b) Suggest how sliding window protocol can improve efficiency.
20. **CSMA/CD**
A LAN has a bit rate of 10 Mbps and a maximum cable length of 2 km. Propagation speed is 2×10^8 m/s.
a) Calculate the minimum frame size for CSMA/CD.
b) Explain why small frames are not allowed.
21. **Framing Techniques**
Explain the following framing techniques with diagrams:
I. Character Count
II. Flag Bytes with Byte Stuffing
III. Bit Stuffing
22. Compare the following MAC protocols:
- Pure ALOHA
 - Slotted ALOHA
 - CSMA/CD

Discuss efficiency, collision probability, and delay.

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